KEDGE Business School EBP1 Quantitative Methods 1

Worksheet 6 — Index numbers

Fully worked solutions

Badr ABOUFARASSE, PhD — 2026–2027

1 Core applications

Master the essential definitions and calculations from the course.

Core application — 20 min

1.1 Exercise 1 – Elementary indices, rebasing and properties

A service costs 80 euros in 2023, 92 euros in 2024 and 101.20 euros in 2025. Calculate the three base-100 indices, verify circularity, calculate the reverse index and verify reversibility, rebase the series to 2024=100, and interpret index 126.5 correctly.

Fully worked solution

$$I_{2024/2023} = 115, \qquad I_{2025/2024} = 110, \qquad I_{2025/2023} = 126.5.$$

Circularity:

$$\frac{115 \times 110}{100} = 126.5.$$

Reverse index:

$$I_{2023/2025} = 100\frac{80}{101.20} \approx 79.0514.$$

Reversibility:

$$126.5 \times 79.0514 \approx 10000.$$

Rebasing to 2024=100:

$$I_{2023}^{(2024)} \approx 86.96, \qquad I_{2024}^{(2024)} = 100, \qquad I_{2025}^{(2024)} = 110.$$

Index 126.5 means the 2025 level is 26.5% above 2023, not 126.5% higher.

Core application — 25 min

1.2 Extra 1 — Complete price–quantity decomposition

A purchasing department observes:

Product	p0	q0	p1	q1
A	4	100	5	90
B	10	50	11	60
C	20	20	18	30

1. Calculate the four totals Σp0q0, Σp1q0, Σp0q1 and Σp1q1.
2. Calculate the value, Laspeyres-price, Paasche-price and Fisher-price indices.
3. Interpret the gap between value and price change.

Method hint:

Complete all four cross-value columns before forming any index.

Fully worked solution

Four totals. Σp0q0 = 400+500+400 = **1,300**. Σp1q0 = 500+550+360 = **1,410**. Σp0q1 = 360+600+600 = **1,560**. Σp1q1 = 450+660+540 = **1,650**.

Value index. 1,650/1,300 × 100 = **126.92**.

Laspeyres price. 1,410/1,300 × 100 = **108.46**. **Paasche price.** 1,650/1,560 × 100 = **105.77**. **Fisher price.** $\sqrt{(108.46 \times 105.77)}$ = **107.11**.

Interpretation. Total purchasing value increased by 26.92%, whereas the balanced price increase is about 7.11%. Most of the value rise therefore reflects a positive quantity or mix effect, not prices alone. The formulas do not by themselves establish why quantities changed.

All four aggregates, computed product by product:

Produit / Product	p_0q_0	p_1q_0	p_0q_1	p_1q_1
A	400	500	360	450
B	500	550	600	660
C	400	360	600	540
Σ	1300	1410	1560	1650

Laspeyres = 1410/1300 × 100 = 108.4615 ; Paasche = 1650/1560 × 100 = 105.7692 ; Fisher = $\sqrt{(108.4615 \times 105.7692)}$ = 107.1069.

2 Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice — 35 min

2.1 Exercise 2 – Laspeyres, Paasche, Fisher and factor reversal

	p_0	q_0	p_1	q_1
Product X	4	100	5	90
Product Y	10	40	12	50

Calculate the four aggregates, Laspeyres, Paasche and Fisher price indices, the corresponding quantity indices, the value index, and verify Fisher factor reversal on a base-100 scale.

Fully worked solution

$$\sum p_0q_0 = 800, \quad \sum p_1q_0 = 980, \quad \sum p_0q_1 = 860, \quad \sum p_1q_1 = 1050.$$

Price indices:

$$I_P^L = 122.50, \qquad I_P^P \approx 122.09, \qquad I_P^F \approx 122.30.$$

Quantity indices:

$$I_Q^L = 107.50, \qquad I_Q^P \approx 107.14, \qquad I_Q^F \approx 107.32.$$

Value index:

$$I_V = 131.25.$$

Factor reversal:

$$\frac{I_P^F I_Q^F}{100} \approx 131.25 = I_V.$$

Verification of the four aggregates by product

Product	p_0q_0	p_1q_0	p_0q_1	p_1q_1
X	4×100=400	5×100=500	4×90=360	5×90=450
Y	10×40=400	12×40=480	10×50=500	12×50=600
Σ	800	980	860	1050

$$L_P = 100\frac{980}{800},$$
$$P_P = 100\frac{1050}{860},$$
$$L_Q = 100\frac{860}{800},$$
$$P_Q = 100\frac{1050}{980}.$$

Verify the Fisher identity using unrounded ratios: Fp×Fq/100=100×1050/800=131.25.

Assessment / exam practice — 45 min

2.2 Exercise 4 — Complete four-product indices — exam style

For four products, prices and quantities are observed in two periods. Base 0 = 100.

Produit / Product	p_0	q_0	p_1	q_1
A	53	20	62	25
B	41	28	52	34
C	75	35	82	38
D	25	30	38	33

1. Build the four product columns p_0q_0, p_1q_0, p_0q_1, p_1q_1 and totals.
2. Compute Laspeyres, Paasche and Fisher price and quantity indices.
3. Compute the value index and verify Fisher's factor reversal. Interpret the three indices.

Fully worked solution

Produit / Product	p_0q_0	p_1q_0	p_0q_1	p_1q_1
A	1060	1240	1325	1550
B	1148	1456	1394	1768
C	2625	2870	2850	3116
D	750	1140	825	1254
Σ	5583	6706	6394	7688

The four aggregates are A=Σp_0q_0=5583, B=Σp_1q_0=6706, C=Σp_0q_1=6394, D=Σp_1q_1=7688.

$$L_P = 100B/A = 100(6706/5583) = 120.114634, \quad P_P = 100D/C = 100(7688/6394) = 120.237723.$$

$$L_Q = 100C/A = 100(6394/5583) = 114.526240, \quad P_Q = 100D/B = 100(7688/6706) = 114.643603.$$

$$F_P = \sqrt{L_P P_P} = 120.176163, \quad F_Q = \sqrt{L_Q P_Q} = 114.584907.$$

$$I_V = 100D/A = 100(7688/5583) = 137.703744, \quad \frac{F_P F_Q}{100} = 137.703744.$$

Prices rise by approximately 20.18%, quantities by 14.58%, and total value by 37.70%. Do not add price and quantity changes: their composition is multiplicative.

3 Optional advanced practice

Explore additional interpretations and professional applications.

Advanced practice — 15 min

3.1 Exercise 3 – Chain indices and deflation

Annual link indices are 105, 108 and 97. Calculate the cumulative chain index and interpret it. Explain why annual changes must be compounded rather than added. Then deflate nominal revenue of 2.4 million euros when the price index equals 120.

Fully worked solution

$$I_{3/0}^{chain} = 100(1.05)(1.08)(0.97) = 109.998 \approx 110.$$

The final level is about 10% above the initial level despite the final 3% annual decrease. Successive changes compound multiplicatively.

Deflation:

$$Real\ value = \frac{2.4}{120/100} = 2.0 \text{ million euros.}$$

Compounding table

Period	Factor	Cumulative index
0	1	100
1	1.05	105
2	1.05×1.08=1.134	113.4
3	1.134×0.97=1.09998	109.998

$$I_{3/0} = 100(1.05)(1.08)(0.97) = 109.998.$$

Adding annual percentages (+5+8−3=10) happens to approximate the exact compounded rate (+9.998%). A deflator of 120 means a price factor of 1.20: 2.4/1.20=2.0 million real euros.

Advanced practice — 25 min

3.2 Extra 2 — Rebasing, inflation and real performance

A nominal revenue series is €500k, €560k and €630k. The corresponding price index (year 1 = 100) is 100, 108 and 120.

1. Calculate real revenue in year-1 euros.
2. Calculate nominal and real growth from year 1 to year 3.
3. Rebase the price index so year 2 = 100.
4. Explain the result to a manager who claims that activity grew by 26%.

Method hint:

Deflate by dividing nominal value by index/100. Rebasing changes the reference, not the underlying inflation path.

Fully worked solution

Real revenue. Year 1: 500/1 = **€500k**. Year 2: 560/1.08 = **€518.52k**. Year 3: 630/1.20 = **€525k**.

Growth. Nominal growth is 630/500 − 1 = **26%**. Real growth is 525/500 − 1 = **5%**.

Rebased price index. Divide each value by 108 and multiply by 100: **92.59, 100, 111.11**.

Managerial interpretation. Revenue received in current euros rose by 26%, but after adjusting for the 20% rise in the price level, purchasing-power-equivalent revenue rose by only 5%. Rebasing to year 2 changes the displayed levels, not this economic conclusion. The selected deflator must match the firm's product mix and coverage.